

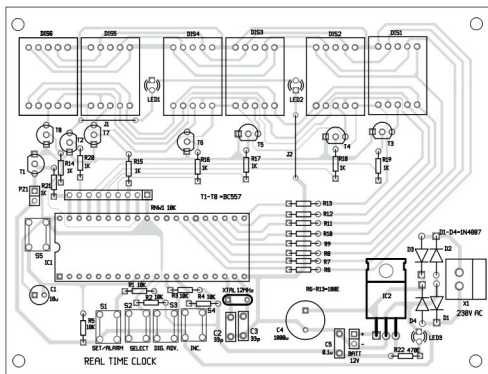


Fig. 5: Component layout for the PCB

(DDMMYY) on the display. Increment the blinking digit number by pressing switch S4. Press S3 to advance to the next digit. Press S3 after setting all the digits of date. The display will show 'Time.' You can set time in the same way as date by using switches S3 and S4. Press S3 after setting all the digits of time. Now the display will show current time.

To set the alarm, press switch S1 twice. The display will show 'Alarm.' Press switch S2 once for 'on' and then press switch S3. The display will show '0000' (HHMM). Set the alarm time by using switches S3 and S4 in the same way as when setting the time. The date can be viewed by pressing any of the keys except S1 and S5.

Functions of All Switches

S1		S2		S3	S4	S5
Switch pressed	Display shows	Switch pressed	Display shows	Digit advance	Increment the blinking digit number	Reset
Once	Set	Once	Date			
		Twice	Time			
Twice	Alarm	Once	On			
		Twice	Off			

The table shows the functions of switches S1 through S5. To start the date and time setting, press

S1. The display will show 'Set.' Now press S2 so that the display shows 'Date.' Press S3 to get '000000'

Software

The software for the real-time clock is written in 'C' language and compiled using the Keil μ Vision4 compiler. Burn the generated hex code into the microcontroller by using a suitable programmer. All the switches are regularly scanned. When any switch is pressed, corresponding action is performed to update the display as per the table.

The author is I/C head (coordinator) in Physics Department of Hemchandracharya North Gujarat University, Patan (N.G.)

CLOCK.C

```
#include <AT89X52.h>
unsigned char bdata scan;
sbit scan_0 = scan ^ 0;
sbit scan_1 = scan ^ 1;
sbit scan_2 = scan ^ 2;
sbit scan_3 = scan ^ 3;
sbit scan_4 = scan ^ 4;
sbit scan_5 = scan ^ 5;
sbit scan_6 = scan ^ 6;
sbit scan_7 = scan ^ 7;
unsigned int i=0,j=0;
bit k1,k2,k3,k0,date=0,alarm=0;
unsigned char f024[]={0x88,0x0a,0x4c,0x49,0x2b,0x19,0x18,0xcb,0x08,0x09,0xa0,0x38,0x9c,0x68,0x1c,0x1e,0xf1,0x3c,0x7e,0x7a}; unsigned char f133[]={0x11,0x7d,0x23,0x29,0x4d,0x89,0x81,0x3d,0x01,0x09,0x05,0xc1,0x93,0x61,0x83,0x87,0xf1,0xc3,0xe7,0xe5};
unsigned char disp1[]={0x88,0x11,0x88,0x11,0x88,0x11,0x88,0x01,0x88,0xe7,0xe5};
unsigned char hh=0,mm=0,sh=0,dh=01,moh=01,yh=01,led=0xff,k=0;
unsigned char ah=0,amh=0;
unsigned char bl=0x01,f[6];
void display(unsigned char,unsigned char,unsigned char,unsigned char,unsigned char);
void time(unsigned char,unsigned char,unsigned char,unsigned char,unsigned char);
```

```
char,unsigned char,unsigned char);
void main(void)
{
    unsigned char
    a=0,b=0,c=0,d=0,temp;
    scan=0xf7;
    TMOD = 0x11;
    TH0 = 0xf8;
    TL0 = 0xf2;
    TH1=0x3c;
    TL1=0xc3;
    TR0 = 1;
    TR1=1;
    ET0=1;
    ET1=1;
    EA=1;
    for(j;j)
    {
        if(alarm==1 && ahb=hh
        && amh=mm)
        {
            scan_0=0;
            if (date==1 && a==0)
            {
                time(hh,mm,sh,led);
                if (temp==sh-2)
                date=0;
            }
            else if(c!=0)
            {
                display(f[0],
                if(j==0)
                {
                    f[0]=temp;
                    else if(j!=0)
                    f[0]=blank;
                }
                else if(date==0 && a==0)
                time(hh,mm,sh,led);
                if (k==0 && (P3 &
                0x0f)!=0x0f)
                {
                    k=1;
                    k0=P3_0;
                    k1=P3_1;
                    k2=P3_2;
                    k3=P3_3;
                }
                if(k==10 && (P3 & 0xf0)
                !=0xf0)
                {
                    k=0;
                    if(alarm==1 &&
                    scan_0==0)
                    {
                        alarm=0;
                        scan_0=1;
                        goto loop1;
                    }
                }
            }
        }
    }
}
```